

Software Engineer – Intern – Mobile Application Assignment Guidelines (2025)

We've designed this test to evaluate your ability to build a mobile application that reflects real-world development scenarios across Android, iOS, and cross-platform mobile development using Flutter. By completing it, you'll demonstrate your understanding of mobile fundamentals, UI/UX principles, and the integration of core features like CRUD operations, search, and filtering. This is your opportunity to showcase your coding standards and creative approach to solving practical problems.

Test Details

You are required to develop a simple mobile application using one of the following technologies:

- Android (Kotlin or Java)
- iOS (Swift – SwiftUI or UIKit)
- Flutter (Dart – cross-platform mobile application development framework for Android and iOS)

The application must include the following features:

- **CRUD Operations**

Implement Create, Read, Update, and Delete functionalities for the main data entity.

- **Search Functionality**

Allow users to search items by name.

- **Filtering**

Allow users to filter items based on selected criteria (e.g., department, category, or any relevant field based on your scenario).

- **Authentication**

Login and registration features using mock APIs or local storage.

- **UI**

You are free to design the UI using platform-appropriate tools:

Android: Jetpack Compose or XML

iOS: SwiftUI or UIKit

Flutter: Flutter widgets using Dart

Tools and Flexibility

You may use any suitable networking library:

- Android: Retrofit / OkHttp
- iOS: URLSession / Alamofire
- Flutter: Dio / http package

If you encounter issues creating or integrating mock APIs, you are allowed to complete the task using alternatives such as Firebase services or SQLite/local storage solutions.

 However, we prioritize proper API integration in this test. Implementing full and correct API functionality will positively influence your score.

API Suggestions (Mock)

MockAPI.io

Reqres.in

UI Design Notes

You are free to design the UI in any way you prefer — creativity is encouraged.

Ensure screens are responsive and scrollable where necessary, especially on smaller devices or when content overflows.

Timeline

Test Start Date: The date you received the test

Time Allocation: 2 days

Submission Deadline: Submit within two days from the assigned date unless an extension has been granted

Submission Process

Please follow these steps:

Push your completed project to a GitHub repository.

Include a README.md file containing:

Brief project overview

Setup and run instructions

API endpoints used

Mention the chosen technology (Android / iOS / Flutter)

UI approach used (Compose/XML, SwiftUI/UIKit, Flutter widgets)

Generate and share:

GitHub repository link

Installable build:

- Android: APK file
- iOS: A screen recorded video
- Flutter: Android APK and/or iOS screen recorded video

Evaluation Criteria

Your submission will be evaluated based on:

Clean Code: Readable, modular, and maintainable structure

UI/UX: Intuitive and user-friendly interface design

Functionality: Proper implementation of all required features

API Integration: Correct usage and handling of mock APIs

Error Handling: Proper handling of network failures, empty states, and invalid inputs

Important Notices

Optional but recommended enhancements:

Display logged-in user details

Logout functionality

Dark/Light mode support

Offline mode support

Use of architectural patterns (e.g., MVVM, Clean Architecture)

Make sure your application builds successfully and the submitted build is installable without errors.